Beyond silico surprisal: Frequency-dependent prediction error updating in speech comprehension

Language is a complex, dynamic sequence, yet the human brain processes it effortlessly. A prominent account of this capacity is predictive processing: the language system uses statistical regularities to predict upcoming words and encodes only the prediction error when the next word occurs. Surprisal, defined as the negative log probability of the next word, is commonly used to quantify prediction error and has been proposed as a causal bottleneck mediating all effects on predictive representations (Levy, 2008). Consistent with this view, many studies report robust surprisal response on behavioral and neural measures, and these surprisal responses may dissociate with other linguistic features, e.g., frequency (Shain, 2024; Staub, 2025; but see Dambacher et al., 2006).

We hypothesize that extensive language exposure induces frequency-dependent efficiencies in lexical representation and access that shape prediction error updating. Specifically, for words matched in surprisal, higher-frequency words may elicit (1) reduced surprisal response because they rely on lower-cost representations (efficient representation), and/or (2) earlier surprisal response because their access and associated error signals propagate more rapidly (efficient propagation) (Fig. 1).

We test these hypotheses in natural speech processing using four large-scale neural datasets (~150,000 words) spanning ECoG and MEG recordings of English- and Mandarin-language narratives. Across datasets, we replicate a classical N400-like response to surprisal. Critically, lexical frequency modulates this response: low-frequency words show amplified surprisal response, and the lowest-frequency words additionally exhibit delayed peak latencies (Fig. 2). This joint pattern supports the coexistence of efficient representation and efficient propagation mechanisms. Moreover, a recent predictive-coding model (Nour Eddine et al., 2024) reproduces these effects when augmented with frequency-dependent representation and propagation efficiencies.

Together, these findings indicate frequency-based efficiencies in prediction-error updating, suggesting that surprisal alone does not fully account for predictive processing. The convergent evidence across languages and neural modalities highlights the value of large, naturalistic datasets for testing mechanistic theories of language processing.

Reference

Dambacher et al., (2006), Frequency and predictability effects on event-related potentials during reading.

Levy (2008), Expectation-based syntactic comprehension.

Eddine et al., (2024), A predictive coding model of the N400.

Shain (2024), Word Frequency and Predictability Dissociate in Naturalistic Reading.

Staub (2025), Predictability in Language Comprehension: Prospects and Problems for Surprisal.

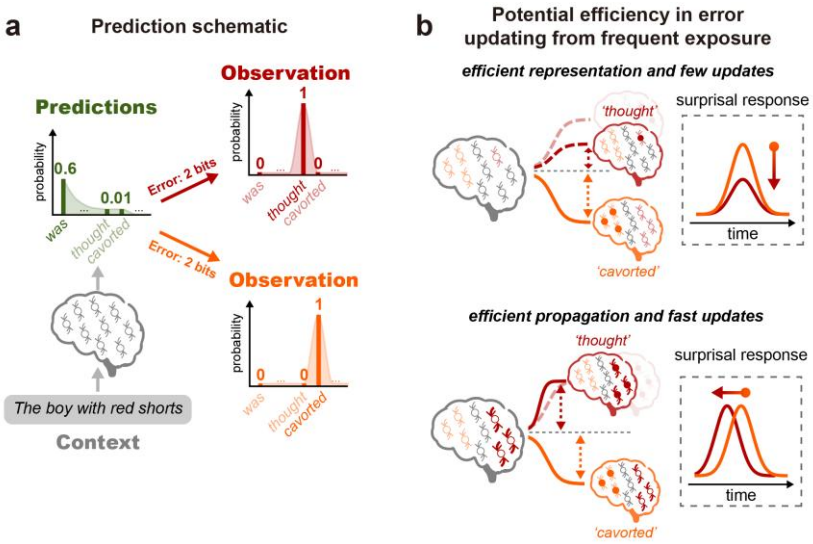

Fig 1. (**a**) A high-frequency word ('thought') and a low-frequency word ('cavorted') can have similar surprisal given a context. (**b**) Hypothesized frequency-based efficiencies.

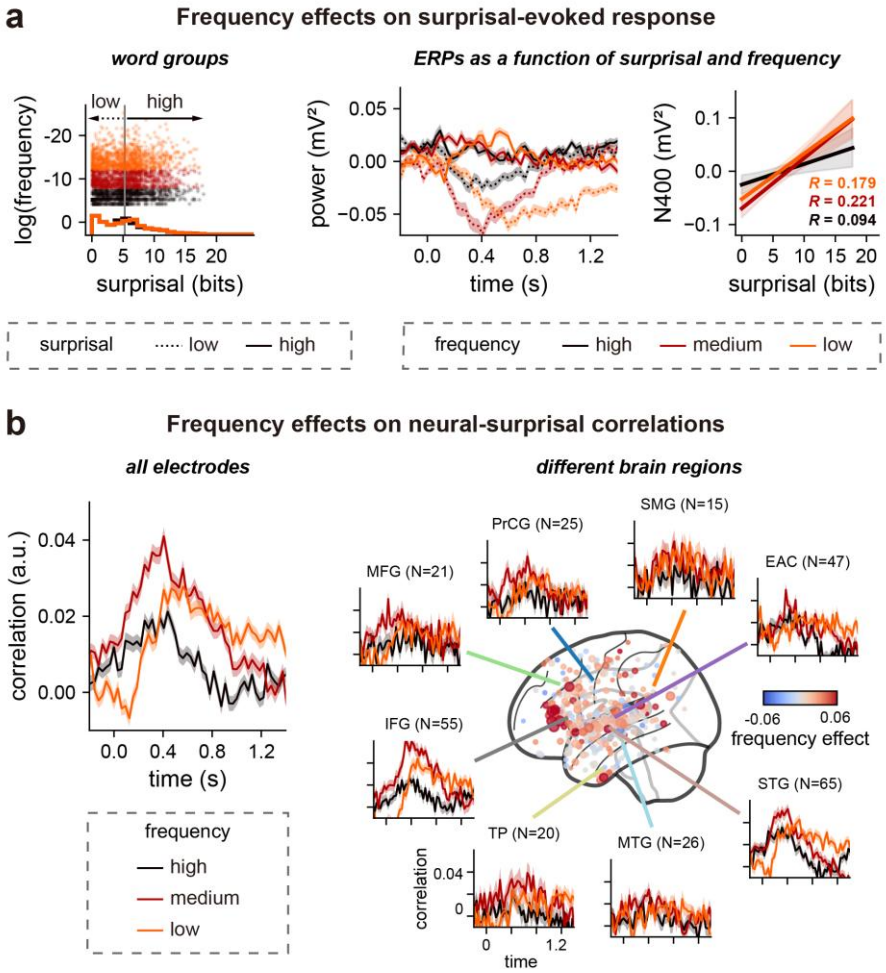

Fig 2. Modulation of the surprisal-related neural response by lexical frequency.